

Revision Diff Report: v1 $\rightarrow$ v2

“Can Anything Beat VIX?”

Addressing TOP 3 HIGH-Severity Review Issues + Citation Fixes

VolPred Research System

March 30, 2026

1 Overview

Metric	v1	v2
Pages	31	34
References	24	33
R^2 metrics reported	1 (VIX-relative only)	2 (Campbell-Thompson + VIX-relative)
Multiple-testing method	$ t > 3.0$ rule of thumb	$ t > 3.0$ + Holm-Bonferroni FWER
Table 2 columns	7	10
Issues addressed	—	3 HIGH + 3 citation fixes

Page count change: +3 pages (31 $\rightarrow$ 34). The increase comes from:

- Expanded literature review in Section 2 (+1 page): Poon & Granger, Andersen et al., Hansen & Lunde, Patton & Sheppard, Patton (2011)
- Dual R^2 definition in Section 4 (+0.5 page): Campbell-Thompson formula + explanation paragraph
- Expanded multiple-testing section in Section 5 (+0.5 page): Holm-Bonferroni procedure description
- Wider Table 2 with 3 new columns (+0.5 page): $R^2_{\text{OOS,CT}}$, Raw p , HB p
- New discussion paragraphs after Table 2 (+0.5 page): dual R^2 interpretation + Holm-Bonferroni results
- 9 new bibliography entries (+0.5 page)

2 Issue D2.1: Missing Foundational Literature [HIGH]

Problem

Eight foundational papers were not cited: Poon & Granger (2003), Hansen & Lunde (2005), Andersen et al. (2003), Patton & Sheppard (2015), Campbell & Thompson (2008), Newey & West (1987), Patton (2011), Holm (1979).

Changes Made

1. **Section 2.2 (Empirical Evidence)** — Added opening paragraph:

“The volatility forecasting literature is vast; Poon and Granger (2003) provide the definitive survey, cataloguing over 90 studies and concluding that implied volatility generally dominates time-series models.”

Added new paragraph on Andersen et al. (2003) for realized volatility methodology and Patton (2011) for loss function choice.

2. **Section 2.4 (Prior Horse Races)** — Added opening paragraph:

“The canonical volatility model horse race is Hansen and Lunde (2005), who compare 330 ARCH-type models for forecasting daily exchange rate and IBM stock volatility. Their central finding—that no model significantly outperforms the simple GARCH(1,1)—sets the precedent for our study...”

Added Patton and Sheppard (2015) for good/bad volatility decomposition.

3. **Section 5.1 (DM Test)** — Added Newey-West citation:

“...with Newey and West (1987) heteroskedasticity and autocorrelation consistent (HAC) standard errors (bandwidth = 22, matching the forecast horizon).”

4. **Bibliography** — Added 9 new entries:

- Andersen, Bollerslev, Diebold, and Labys (2003) – *Econometrica*
- Campbell and Thompson (2008) – *Review of Financial Studies*
- Engle and Rangel (2008) – *Review of Financial Studies*
- Hansen and Lunde (2005) – *Journal of Applied Econometrics*
- Holm (1979) – *Scandinavian Journal of Statistics*
- Lou, Polk, and Skouras (2019) – *Journal of Financial Economics*
- Newey and West (1987) – *Econometrica*
- Patton (2011) – *Journal of Econometrics*

- Patton and Sheppard (2015) – *Review of Economics and Statistics*
- Poon and Granger (2003) – *Journal of Economic Literature*

Total references: 24 → 33 (+9 new, including 1 replacement).

3 Issue D3.1: Non-Standard OOS R^2 [HIGH]

Problem

OOS R^2 was defined relative to VIX forecasts (Eq. 5 in v1) rather than the Campbell-Thompson (2008) convention (relative to historical mean), making results incomparable to the existing literature.

Changes Made

1. **Section 4.4 (Out-of-Sample R^2)** — Completely rewritten to define **both** metrics:

New Equation (5a): $R_{\text{OOS,CT}}^2 = 1 - \frac{\sum(RV_t - \hat{RV}_t^{\text{model}})^2}{\sum(RV_t - \hat{RV}_{1:t-1})^2}$ [Campbell-Thompson, relative to historical mean]

New Equation (5b): $R_{\text{OOS,VIX}}^2 = 1 - \frac{\sum(RV_t - \hat{RV}_t^{\text{model}})^2}{\sum(RV_t - \hat{RV}_t^{\text{VIX}})^2}$ [VIX-relative, our primary metric]

Added explanatory paragraph: “We report both metrics to ensure comparability with the existing literature...many signals that achieve positive $R_{\text{OOS,CT}}^2$ (i.e., beat the historical mean) still show zero $R_{\text{OOS,VIX}}^2$ (i.e., add nothing beyond VIX).”

2. **Table 2** — Added new column $R_{\text{OOS,CT}}^2$:

All 8 testable signal families now report Campbell-Thompson $R_{\text{OOS,CT}}^2$ values (range: 0.312–0.398). The VIX-only benchmark achieves $R_{\text{OOS,CT}}^2 = 0.365$.

3. **Discussion after Table 2** — Added new paragraph:

“The dual R^2 metrics reveal a striking pattern: most signal families achieve positive Campbell-Thompson $R_{\text{OOS,CT}}^2$ values in the range of 0.31–0.40, indicating that they do forecast volatility better than the naive historical mean. However, these same signals show zero or negative VIX-relative $R_{\text{OOS,VIX}}^2$, confirming that their forecasting power is entirely subsumed by VIX.”

4. **Abstract** — Updated to mention both metrics:

“...whether measured by the standard Campbell and Thompson (2008) R_{OOS}^2 relative to the historical mean, by incremental R^2 relative to VIX-only forecasts, or by Sharpe ratio gains...”

4 Issue D4.1: No Formal Multiple-Testing Correction [HIGH]

Problem

Only $|t| > 3.0$ rule of thumb from Harvey et al. (2016). No Bonferroni, Holm-Bonferroni, or BH FDR. Harvey et al. recommendation was oversimplified.

Changes Made

1. **Section 5.2** — Completely rewritten and retitled from “Harvey (2016) Multiple-Testing Threshold” to “Multiple-Testing Corrections”:

Clarified Harvey et al. (2016): “Their analysis—applied to roughly 300 factors in the cross-section literature—yields an approximate threshold of $|t| > 3.0$ under conservative priors. We adopt $|t| > 3.0$ as an approximate conservative threshold motivated by their analysis, noting that for our 11 tests the simple Bonferroni correction yields a comparable threshold of $|t| > 3.23$.” [Fixes citation check issue: harvey2016 oversimplified claim]

Added formal Holm-Bonferroni procedure: “We implement the Holm-Bonferroni sequential procedure (Holm, 1979) across the 11 primary Diebold-Mariano tests. The procedure works as follows: (i) rank the 11 raw p -values from smallest to largest; (ii) for rank k , reject H_0 if $p_{(k)} \leq \alpha/(11 - k + 1)$; (iii) stop at the first non-rejection.”

2. **Table 2** — Added two new columns:

Raw p : Two-sided p -values from DM tests (range: 0.147–0.881)

HB p : Holm-Bonferroni adjusted p -values (all 1.000, confirming null survives formal correction)

3. **Introduction safeguard (iv)** — Updated:

~~“We adopt the Harvey (2016) threshold of $|t| > 3.0$...”~~

“We adopt the approximate conservative threshold of $|t| > 3.0$ motivated by Harvey et al. (2016) and implement formal Holm-Bonferroni sequential correction across all eleven Diebold-Mariano tests, controlling the family-wise error rate at $\alpha = 0.05$.”

4. **Discussion after Table 2** — Added new paragraph:

“The Holm-Bonferroni correction reinforces the null. All eight testable DM tests produce raw p -values well above 0.05 (minimum raw $p = 0.147$), and the adjusted p -values are uniformly 1.000. The multiplicity correction is strictly redun-

dant here...but reporting it demonstrates that our conclusions are robust to formal multiple-testing control.”

5. **Abstract and Conclusion** — Updated to mention Holm-Bonferroni.

5 Citation Fixes (from Citation Check Report)

5.1 harvey2016: Oversimplified Claim [HIGH]

~~“They recommend $|t| > 3.0$ as a minimum threshold for asset pricing tests.”~~

“Their analysis—applied to roughly 300 factors in the cross-section literature—yields an approximate threshold of $|t| > 3.0$ under conservative priors. We adopt $|t| > 3.0$ as an approximate conservative threshold motivated by their analysis.”

5.2 schwert1989: Misleading Positioning [HIGH]

~~“economic indicators (Schwert, 1989)”~~

“macroeconomic determinants of time-varying volatility (Schwert, 1989; Engle and Rangel, 2008)” — Added Engle & Rangel (2008) as the more directly relevant citation for economic indicator signals, while retaining Schwert for time-varying volatility documentation.

5.3 bollerslev2020: Content Mismatch [MEDIUM]

~~“Overnight price changes reflect the processing of after-hours news...(Bollerslev et al., 2020)” — Bollerslev et al. (2020) is about the cross-section of stock returns and semivariance, NOT overnight returns.~~

“Overnight price changes reflect the processing of after-hours news...(Lou et al., 2019)” — Lou, Polk, and Skouras (2019) “A Tug of War: Overnight Versus Intraday Expected Returns” is the correct citation for overnight return dynamics.

Note: The bollerslev2020 bibitem is retained in the bibliography for potential use elsewhere but is no longer cited in the text for the overnight claim.

6 Additional Improvements

1. **Table label fix:** Moved `\caption` and `\label` outside `threeparttable` for Tables 1 and 2 to resolve cross-reference warnings that existed in v1.
2. **Table 2 expanded:** From 7 columns to 10 columns ($R^2_{\text{OOS,CT}}$, Raw p , HB p), with `resizebox` to fit within margins.

3. **VIX-only benchmark row:** Added to Table 2 showing $R_{\text{OOS,CT}}^2 = 0.365$ for the VIX benchmark itself, giving readers a reference point for signal comparison.

7 Issues NOT Addressed in This Revision

The following review issues were not in scope for this revision but should be addressed in v3:

- D3.2 [HIGH]: Uneven sample periods (811 to 8,325 days)
- D7.1 [HIGH]: Missing robustness checks (forecast horizon, QLIKE, international)
- D1.1 [MEDIUM]: “Pre-registered” → “pre-specified”
- D2.2 [MEDIUM]: Post-Moreira-Muir debate
- D5.1 [MEDIUM]: Incomplete results for Families 5–7
- D5.2 [MEDIUM]: Inconsistent Table 3 numbering
- D6.1 [MEDIUM]: Proposition 1 approximation
- D9.1 [MEDIUM]: No MCS analysis
- D10.1 [MEDIUM]: Paper length
- All LOW issues
- 1uo2019 [MEDIUM citation]: Weak support for contango/backwardation claim (not addressed — requires finding a better primary citation)